

OSSP Practical 5,6

Practical 5

Code:

```
polisettysreeharsha@SREEHA x + v
GNU nano 8.7.1 practical5.c
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    int fd[2];

    pipe(fd);

    if(fork() == 0)
    {
        dup2(fd[1], STDOUT_FILENO);
        close(fd[0]);
        close(fd[1]);
        execlp("ls", "ls", "-l", NULL) ;
    }
    if(fork() == 0)
    {
        dup2(fd[0], STDIN_FILENO);
        close(fd[0]);
        close(fd[1]);
        execlp("grep", "grep", ".c", NULL) ;
    }
    close(fd[0]);
    close(fd[1]);

    wait(NULL);
    wait(NULL);

    return 0;
}
```

Output:

```
polisettyreearsha@SREEHARSHA:~$ nano practical5.c
polisettyreearsha@SREEHARSHA:~$ gcc practical5.c -o practical5
polisettyreearsha@SREEHARSHA:~$ ./practical5
-rwxr-xr-x 1 polisettyreearsha polisettyreearsha 15992 Aug 12 16:51 Exclp
-rw-r--r-- 1 polisettyreearsha polisettyreearsha 163 Aug 12 16:51 Exclp.c
-rw-r--r-- 1 polisettyreearsha polisettyreearsha 71 Aug 1 06:07 Execute.c.save
-rwxr-xr-x 1 polisettyreearsha polisettyreearsha 16104 Sep 2 04:52 FCFSschedule
-rwxr-xr-x 1 polisettyreearsha polisettyreearsha 456 Sep 2 04:55 FCFSschedule.c
-rw-r--r-- 1 polisettyreearsha polisettyreearsha 242 Aug 12 16:58 Orphan.c
-rw-r--r-- 1 polisettyreearsha polisettyreearsha 1044 Sep 2 05:13 Roundrobin.c
-rw-r--r-- 1 polisettyreearsha polisettyreearsha 542 Sep 2 05:04 SJFnonpreemptive.c
-rwxr-xr-x 1 polisettyreearsha polisettyreearsha 16400 Aug 26 05:16 Skill3code
-rw-r--r-- 1 polisettyreearsha polisettyreearsha 1001 Aug 26 05:16 Skill3code.c
-rwxr-xr-x 1 polisettyreearsha polisettyreearsha 16384 Aug 26 05:23 Skill4code
-rw-r--r-- 1 polisettyreearsha polisettyreearsha 745 Aug 26 05:22 Skill4code.c
-rwxr-xr-x 1 polisettyreearsha polisettyreearsha 16088 Aug 12 06:43 childbyfork
-rw-r--r-- 1 polisettyreearsha polisettyreearsha 225 Aug 12 06:43 childbyfork.c
-rwxr-xr-x 1 polisettyreearsha polisettyreearsha 16088 Aug 12 16:54 childbyparconc
-rw-r--r-- 1 polisettyreearsha polisettyreearsha 156 Aug 12 16:53 childbyparconc.c
-rwxr-xr-x 1 polisettyreearsha polisettyreearsha 16176 Aug 6 04:13 copy
-rwxr-xr-x 1 polisettyreearsha polisettyreearsha 16048 Aug 27 04:10 fifo_create
-rw-r--r-- 1 polisettyreearsha polisettyreearsha 177 Aug 27 03:58 fifo_create.c
-rw-r--r-- 1 polisettyreearsha polisettyreearsha 214 Aug 27 04:05 fifo_read.c
-rw-r--r-- 1 polisettyreearsha polisettyreearsha 251 Aug 27 04:10 fifo_write.c
-rw-r--r-- 1 polisettyreearsha polisettyreearsha 71 Jul 29 05:55 hello.c
-rwxr-xr-x 1 polisettyreearsha polisettyreearsha 16048 Aug 12 16:49 multiplechildproc
-rw-r--r-- 1 polisettyreearsha polisettyreearsha 168 Aug 12 16:49 multiplechildproc.c
-rwxr-xr-x 1 polisettyreearsha polisettyreearsha 16000 Aug 1 05:44 parent_child
-rw-r--r-- 1 polisettyreearsha polisettyreearsha 175 Aug 1 05:42 parent_child.c
-rwxr-xr-x 1 polisettyreearsha polisettyreearsha 16048 Aug 1 05:52 parent_child2
-rw-r--r-- 1 polisettyreearsha polisettyreearsha 215 Aug 1 05:51 parent_child2.c
-rw-r--r-- 1 polisettyreearsha polisettyreearsha 420 Aug 27 03:31 pipe1.c
-rwxr-xr-x 1 polisettyreearsha polisettyreearsha 16208 Aug 18 14:28 prac3
-rw-r--r-- 1 polisettyreearsha polisettyreearsha 310 Aug 18 14:27 prac3.c
-rwxr-xr-x 1 polisettyreearsha polisettyreearsha 16120 Aug 18 14:33 prac4
-rw-r--r-- 1 polisettyreearsha polisettyreearsha 259 Aug 18 14:33 prac4.c
-rwxr-xr-x 1 polisettyreearsha polisettyreearsha 16312 Aug 5 08:23 practical1
-rw-r--r-- 1 polisettyreearsha polisettyreearsha 441 Aug 5 08:23 practical1.c
-rwxr-xr-x 1 polisettyreearsha polisettyreearsha 16176 Aug 6 03:56 practical2
-rw-r--r-- 1 polisettyreearsha polisettyreearsha 446 Aug 5 08:33 practical2.c
-rwxr-xr-x 1 polisettyreearsha polisettyreearsha 16216 Sep 9 05:24 practical5
```

-rw-r--r--	1	polisettyreearsha	polisettyreearsha	420	Aug	27	03:31	pipe1.c
-rwxr-xr-x	1	polisettyreearsha	polisettyreearsha	16208	Aug	18	14:28	prac3
-rw-r--r--	1	polisettyreearsha	polisettyreearsha	310	Aug	18	14:27	prac3.c
-rwxr-xr-x	1	polisettyreearsha	polisettyreearsha	16120	Aug	18	14:33	prac4
-rw-r--r--	1	polisettyreearsha	polisettyreearsha	259	Aug	18	14:33	prac4.c
-rwxr-xr-x	1	polisettyreearsha	polisettyreearsha	16312	Aug	5	08:23	practical1
-rw-r--r--	1	polisettyreearsha	polisettyreearsha	441	Aug	5	08:23	practical1.c
-rwxr-xr-x	1	polisettyreearsha	polisettyreearsha	16176	Aug	6	03:56	practical2
-rw-r--r--	1	polisettyreearsha	polisettyreearsha	446	Aug	5	08:33	practical2.c
-rwxr-xr-x	1	polisettyreearsha	polisettyreearsha	16216	Sep	9	05:24	practical5
-rw-r--r--	1	polisettyreearsha	polisettyreearsha	383	Sep	9	05:24	practical5.c
-rw-r--r--	1	polisettyreearsha	polisettyreearsha	1048	Sep	2	05:10	priority.c
-rw-----	1	polisettyreearsha	polisettyreearsha	19	Jul	29	04:57	process.c.save
-rwxr-xr-x	1	polisettyreearsha	polisettyreearsha	16000	Aug	12	06:36	processid
-rw-r--r--	1	polisettyreearsha	polisettyreearsha	111	Aug	12	06:36	processid.c
-rwxr-xr-x	1	polisettyreearsha	polisettyreearsha	16080	Aug	12	16:46	pwaitc
-rw-r--r--	1	polisettyreearsha	polisettyreearsha	265	Aug	12	16:46	pwaitc.c
-rwxr-xr-x	1	polisettyreearsha	polisettyreearsha	16200	Sep	2	05:47	sigchild
-rw-r--r--	1	polisettyreearsha	polisettyreearsha	541	Sep	2	05:47	sigchild.c
-rw-r--r--	1	polisettyreearsha	polisettyreearsha	260	Aug	29	05:14	signal.c
-rw-r--r--	1	polisettyreearsha	polisettyreearsha	513	Sep	2	05:50	siggpipe.c
-rw-r--r--	1	polisettyreearsha	polisettyreearsha	413	Sep	2	05:51	sigsegmentationv.c
-rw-r--r--	1	polisettyreearsha	polisettyreearsha	281	Sep	2	05:31	sigterm.c
-rwxr-xr-x	1	polisettyreearsha	polisettyreearsha	16192	Aug	26	05:07	simplecommandloop
-rw-r--r--	1	polisettyreearsha	polisettyreearsha	310	Aug	26	05:06	simplecommandloop.c
-rwxr-xr-x	1	polisettyreearsha	polisettyreearsha	16184	Sep	3	04:01	socketclient
-rw-r--r--	1	polisettyreearsha	polisettyreearsha	564	Sep	3	03:49	socketclient.c
-rwxr-xr-x	1	polisettyreearsha	polisettyreearsha	16352	Sep	3	04:01	socketserver
-rw-r--r--	1	polisettyreearsha	polisettyreearsha	691	Sep	3	03:40	socketserver.c
-rw-r--r--	1	polisettyreearsha	polisettyreearsha	689	Sep	3	04:00	socketserver.c.save
-rw-r--r--	1	polisettyreearsha	polisettyreearsha	211	Aug	1	05:57	wait.c
-rw-r--r--	1	polisettyreearsha	polisettyreearsha	193	Aug	22	04:47	zombie.c

Practical 6:

Code:

```

GNU nano 8.7.1 practical6.c
#include <stdio.h>
#include <signal.h>
#include <unistd.h>

void handler(int sig)
{
    if(sig == SIGINT)
        printf("\nSIGINT Received\n");

    if(sig == SIGTERM)
        printf("\nSIGTERM Received\n");

    if(sig == SIGUSR1)
        printf("\nSIGUSR1 Received\n");
}

int main()
{
    signal(SIGINT, handler);
    signal(SIGTERM, handler);
    signal(SIGUSR1, handler);

    while(1)
    {
        printf("Running ... \n");
        sleep(3);
    }
    return 0;
}

```

Output:

```
polisettysreeharsha@SREEHARSHA:~$ nano practical6.c
polisettysreeharsha@SREEHARSHA:~$ gcc practical6.c -o practical6
polisettysreeharsha@SREEHARSHA:~$ ./practical6
Running ...
Running ...
Running ...
^C
SIGINT Received
Running ...
kRunning ...
Running ...
killRunning ...
Running ...
Running ...
Running ...
Running ...
Running ...
kill -SIGUSR1
Running ...
^X^Z
[1]+  Stopped                  ./practical6
```